Erlang Age-Transport Dynamics:
Algebraic Renewal Roots versus the Essential Spectral Edge

Route-A source-local certificate HCS-C272

1 September 2026

Abstract

We solve an Erlang-fertility McKendrick age-transport semigroup at the level of characteristics, renewal poles, and long-time dynamics. The Euler–Lotka denominator has exactly k explicit algebraic roots. A rank-one resolvent comparison fixes the essential edge at $-\mu$ and proves that a formal root is an L^1 eigenvalue only when it lies strictly to the right of that edge. For $\beta > 1$ the dominant real root owns a positive asynchronous rank-one term, while population growth starts only at $\beta = (1 + \mu/\gamma)^k$; 2,340 root cells audit the distinction. No renewal pole is promoted to target arithmetic data.

1 Erlang age-transport owner

On $X = L^1(\mathbb{R}_+, da)$, freeze

$$\partial_t n + \partial_a n = -\mu n, \qquad n(t,0) = \int_0^\infty b_k(a) n(t,a)\, da, \tag{1}$$

$$b_k(a) = \frac{\beta\gamma^k a^{k-1} e^{-\gamma a}}{(k-1)!}, \quad \mu, \gamma, \beta > 0, \quad k \geq 1. \tag{2}$$

Age-structured transport descends from M'Kendrick's mathematical population framework [1]; the Erlang specialization and spectral audit are local.

Characteristics transport initial mass to the right with factor $e^{-\mu t}$ and fill the newborn triangle from the boundary. Substitution into (1) produces a scalar renewal equation. Its exponential mode has denominator

$$\Delta(\lambda) = 1 - \beta \left(\frac{\gamma}{\gamma + \lambda + \mu} \right)^k. \tag{3}$$

Therefore the algebraic roots are

$$\lambda_j = \gamma\beta^{1/k} e^{2\pi i j/k} - \gamma - \mu, \qquad 0 \leq j < k. \tag{4}$$

2 Resolvent and the eigenvalue gate

Let $Gf = -f' - \mu f$ with the renewal boundary from (1). For $\Re\lambda > -\mu$, direct integration gives

$$u(a) = e^{-(\lambda+\mu)a} u(0) + \int_0^a e^{-(\lambda+\mu)(a-s)} f(s)\, ds. \tag{5}$$

The boundary determines $u(0)$ through (3). Relative to the absorbing mortality shift, the remaining term in (5) is one scalar functional times one exponential; the resolvent difference is rank one. The essential spectral edge is consequently $\Re\lambda = -\mu$.

Theorem 1 (root/eigenvalue firewall)**.** *A root in (4) is an L^1 eigenvalue exactly when $\Re\lambda_j > -\mu$. Roots on or below the essential edge are not L^1 eigenvalues.*

Proof. Every candidate eigenfunction is a multiple of $e^{-(\lambda_j+\mu)a}$. Its absolute integral is finite exactly under the strict displayed inequality. Equation (3) then supplies, and only then supplies, the renewal boundary. □

3 Two distinct transitions

Put $\rho = \beta^{1/k}$. The real root

$$\lambda_0 = \gamma(\rho - 1) - \mu \tag{6}$$

has strictly largest real part. It clears the essential edge exactly when $\beta > 1$. Let $T(t)$ be the renewal semigroup and $S_\mu(t)f(a) = e^{-\mu t} f(a-t)\mathbf{1}_{a\geq t}$. Characteristics show that $T(t) - S_\mu(t)$ is the newborn term. On $0 \leq s \leq t$ its initial-data forcing is

$$g_f(s) = e^{-\mu s} \int_0^\infty b_k(s+u) f(u)\, du.$$

The bounded derivative and decaying tail of b_k make $f \mapsto g_f$ compact from L^1 into $C[0,t]$ by Arzelà–Ascoli. The Volterra renewal resolvent and newborn-triangle map preserve compactness. Thus $T(t) - S_\mu(t)$ is compact, $\|S_\mu(t)\|_{\text{ess}} = e^{-\mu t}$, and the essential growth bound is $-\mu$.

The simple positive residue at λ_0 defines a positive rank-one projection P_0. Quasi-compact spectral decomposition now gives

$$\|e^{-\lambda_0 t} T(t) - P_0\|_{L^1 \to L^1} \longrightarrow 0, \tag{7}$$

with stable-age profile proportional to $e^{-\gamma(\rho-1)a}$. For $k > 1$, its separation from the rest is

$$\min\{\gamma(\rho - 1),\, \gamma\rho[1 - \cos(2\pi/k)]\}; \tag{8}$$

for $k = 1$, only the first term remains. When $\beta \leq 1$, no isolated real pole lies beyond the edge, so we make no universal rank-one claim.

The sign of (6) changes at the later threshold

$$\beta_c = (1 + \mu/\gamma)^k. \tag{9}$$

Thus, for every nonzero positive initial density, $1 < \beta < \beta_c$ has an isolated positive eigenprofile whose amplitude still decays, $\beta = \beta_c$ is population-critical, and $\beta > \beta_c$ grows. For $\beta \leq 1$ the population still decays, but no universal rank-one profile is claimed. At zero fertility, only the mortality-weighted right shift remains.

4 Certificate and Route-A boundary

The exact receipt covers 360 parameter cases, all 2,340 roots for $1 \leq k \leq 12$, exact polynomials, edge labels, gaps, and six zero-birth faces. Independent polynomial/root reconstruction, 48 symbolic identities, fresh byte replay, and repaired-hash hostile tests audit the release. These finite cells are regression evidence; (3)–(9) prove the parameter theorem.

Under `NO_BAD_EULER_OR_ROOT_NUMBER`, the tuple is

```
(A0_FAIL,A1_FAIL,A2_FAIL,A3_FAIL,A4_FORMAL_HINT),
```

overall `ROUTE_A_REJECTED`; Route B is disabled. The source transport generator and its renewal poles are not a target determinant, divisor, functional equation, or Hilbert–Pólya operator.

References

[1] A. G. M'Kendrick, *Applications of Mathematics to Medical Problems*, Proceedings of the Edinburgh Mathematical Society 44 (1925), 98–130, doi:10.1017/S0013091500034428.